

7. OPERATIONS ON THE CARDIOVASCULAR SYSTEM (35-39)

35 Operations on valves and septa of heart

Includes: sternotomy (median) (transverse) as operative approach
thoracotomy as operative approach

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))

35.0 Closed heart valvotomy

Excludes: percutaneous (balloon) valvuloplasty ([35.96](#))

35.00 Closed heart valvotomy, unspecified valve

35.01 Closed heart valvotomy, aortic valve

35.02 Closed heart valvotomy, mitral valve

35.03 Closed heart valvotomy, pulmonary valve

35.04 Closed heart valvotomy, tricuspid valve

35.1 Open heart valvuloplasty without replacement

Includes: open heart valvotomy

Excludes: that associated with repair of:
endocardial cushion defect ([35.54](#), [35.63](#), [35.73](#))
valvular defect associated with atrial and ventricular septal defects ([35.54](#), [35.63](#), [35.73](#))
percutaneous (balloon) valvuloplasty (35.96)

Code also cardiopulmonary bypass if performed [extracorporeal circulation] [heart-lung machine] (39.61)

35.10 Open heart valvuloplasty without replacement, unspecified valve

35.11 Open heart valvuloplasty of aortic valve without replacement

35.12 Open heart valvuloplasty of mitral valve without replacement

35.13 Open heart valvuloplasty of pulmonary valve without replacement

35.14 Open heart valvuloplasty of tricuspid valve without replacement

35.2 Replacement of heart valve

Includes: excision of heart valve with replacement

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] (39.61)

Excludes: that associated with repair of:
endocardial cushion defect ([35.54](#), [35.63](#), [35.73](#))
valvular defect associated with atrial and ventricular septal defects ([35.54](#), [35.63](#), [35.73](#))

35.20 Replacement of unspecified heart valve
Repair of unspecified heart valve with tissue graft or prosthetic implant

35.21 Replacement of aortic valve with tissue graft
Repair of aortic valve with tissue graft (autograft) (heterograft) (homograft)
[Surgical procedure of the Aortic Root with Valve Procedure \(IM\)](#)

- 35.22 Other replacement of aortic valve
 - Repair of aortic valve with replacement:
 - NOS
 - prosthetic (partial) (synthetic) (total)
 - [Pulmonary valve replacement percutaneous \(IM\)](#)
 - [TAVI percutaneous \(IM\)](#)
- 35.23 Replacement of mitral valve with tissue graft
 - Repair of mitral valve with tissue graft (autograft) (heterograft) (homograft)
- 35.24 Other replacement of mitral valve
 - Repair of mitral valve with replacement:
 - NOS
 - prosthetic (partial) (synthetic) (total)
 - [Mitral clipping percutaneous](#)
 - [Mitral valve replacement percutaneous](#)

Excludes: percutaneous repair with implant or leaflet clip ([35.97](#))
- 35.25 Replacement of pulmonary valve with tissue graft
 - Repair of pulmonary valve with tissue graft (autograft) (heterograft) (homograft)
- 35.26 Other replacement of pulmonary valve
 - Repair of pulmonary valve with replacement:
 - NOS
 - prosthetic (partial) (synthetic) (total)
- 35.27 Replacement of tricuspid valve with tissue graft
 - Repair of tricuspid valve with tissue graft (autograft) (heterograft) (homograft)
- 35.28 Other replacement of tricuspid valve
 - Repair of tricuspid valve with replacement:
 - NOS
 - prosthetic (partial) (synthetic) (total)
- 35.3 Operations on structures adjacent to heart valves

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))

 - 35.31 Operations on papillary muscle
 - Division of papillary muscle
 - Reattachment of papillary muscle
 - Repair of papillary muscle
 - 35.32 Operations on chordae tendineae
 - Division of chordae tendineae
 - Repair of chordae tendineae
 - 35.33 Annuloplasty
 - Plication of annulus
 - 35.34 Infundibulectomy
 - Right ventricular infundibulectomy
 - [Resection of Intracardiac Mass](#)
 - 35.35 Operations on trabeculae carneae cordis
 - Division of trabeculae carneae cordis
 - Excision of trabeculae carneae cordis
 - Excision of aortic subvalvular ring

- 35.39 Operations on other structures adjacent to valves of heart
Repair of sinus of Valsalva (aneurysm)
- 35.4 Production of septal defect in heart
 - 35.41 Enlargement of existing atrial septal defect
Rashkind procedure
Septostomy (atrial) (balloon)
[Ballon Atrial Septostomy \(BAS\)](#)
 - 35.42 Creation of septal defect in heart
Blalock-Hanlon operation
- 35.5 Repair of atrial and ventricular septa with prosthesis
Includes: repair of septa with synthetic implant or patch
Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))
 - 35.50 Repair of unspecified septal defect of heart with prosthesis
Excludes: that associated with repair of:
endocardial cushion defect ([35.54](#))
septal defect associated with valvular defect ([35.54](#))
 - 35.51 Repair of atrial septal defect with prosthesis, open technique
Atrioseptoplasty with prosthesis
Correction of atrial septal defect with prosthesis
Repair:
foramen ovale (patent)
with prosthesis ostium secundum defect with prosthesis
Excludes: that associated with repair of:
atrial septal defect associated with valvular and ventricular septal defects ([35.54](#))
endocardial cushion defect ([35.54](#))
 - 35.52 Repair of atrial septal defect with prosthesis, closed technique
Insertion of atrial septal umbrella [King-Mills]
[Atrial Septal Defect closure with device](#)
[Atrial Septal Stenting](#)
[Patent Foramen Ovale closure with device](#)
[Sinus Valsava Repair](#)
 - 35.53 Repair of ventricular septal defect with prosthesis, open technique
Correction of ventricular septal defect with prosthesis
Repair of supracristal defect with prosthesis
Excludes: that associated with repair of:
endocardial cushion defect ([35.54](#))
ventricular defect associated with valvular and atrial septal defects ([35.54](#))
 - 35.54 Repair of endocardial cushion defect with prosthesis
Repair:
atrioventricular canal with prosthesis (grafted to septa)
ostium primum defect with prosthesis (grafted to septa)
valvular defect associated with atrial and ventricular septal defects with prosthesis
(grafted to septa)
Excludes: repair of isolated:
atrial septal defect ([35.51-35.52](#))
valvular defect ([35.20](#), [35.22](#), [35.24](#), [35.26](#), [35.28](#))

ventricular septal defect ([35.53](#))

35.55 Repair of ventricular septal defect with prosthesis, closed technique

VSD closure with device

35.6 Repair of atrial and ventricular septa with tissue graft

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))

35.60 Repair of unspecified septal defect of heart with tissue graft

Excludes: that associated with repair of:

endocardial cushion defect ([35.63](#))

septal defect associated with valvular defect ([35.63](#))

35.61 Repair of atrial septal defect with tissue graft

Atrioseptoplasty with tissue graft

Correction of atrial septal defect with tissue graft

Repair:

foramen ovale (patent) with tissue graft

ostium secundum defect with tissue graft

Excludes: that associated with repair of:

atrial septal defect associated with valvular and ventricular septal defects ([35.63](#))

endocardial cushion defect ([35.63](#))

35.62 Repair of ventricular septal defect with tissue graft

Correction of ventricular septal defect with tissue graft

Repair of supracristal defect with tissue graft

Excludes: that associated with repair of:

endocardial cushion defect ([35.63](#))

ventricular defect associated with valvular and atrial septal defects ([35.63](#))

35.63 Repair of endocardial cushion defect with tissue graft

Repair of:

atrioventricular canal with tissue graft

ostium primum defect with tissue graft

valvular defect associated with atrial and ventricular septal defects with tissue graft

Excludes: repair of isolated:

atrial septal defect ([35.61](#))

valvular defect ([35.20-35.21](#), [35.23](#), [35.25](#), [35.27](#))

ventricular septal defect ([35.62](#))

35.7 Other and unspecified repair of atrial and ventricular septa

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))

35.70 Other and unspecified repair of unspecified septal defect of heart

Repair of septal defect NOS

Excludes: that associated with repair of:

endocardial cushion defect ([35.73](#))

septal defect associated with valvular defect ([35.73](#))

35.71 Other and unspecified repair of atrial septal defect

Repair NOS:

atrial septum
foramen ovale (patent)
ostium secundum defect

[Replacement of unspecified heart valve with Sinus Valsava Repair](#)

Excludes: that associated with repair of:

atrial septal defect associated with valvular ventricular septal defects ([35.73](#))
endocardial cushion defect ([35.73](#))

35.72 Other and unspecified repair of ventricular septal defect

Repair NOS:

supracristal defect
ventricular septum

Repair of :

[Ventricular Septal defect with prosthesis open technique with Sinus Valsava Repair](#)
[Ventricular septal defect and resection and anastomosis of aorta](#)

Excludes: that associated with repair of:

endocardial cushion defect ([35.73](#))
ventricular septal defect associated with valvular and atrial septal defects ([35.73](#))

35.73 Other and unspecified repair of endocardial cushion defect

Repair NOS:

atrioventricular canal
ostium primum defect
valvular defect associated with atrial and ventricular septal defects

Excludes: repair of isolated:

atrial septal defect ([35.71](#))
valvular defect ([35.20](#), [35.22](#), [35.24](#), [35.26](#), [35.28](#))
ventricular septal defect ([35.72](#))

35.8 Total repair of certain congenital cardiac anomalies

Note: For partial repair of defect [e.g. repair of atrial septal defect in tetralogy of Fallot] -- code to specific procedure

35.81 Total repair of tetralogy of Fallot

One-stage total correction of tetralogy of Fallot with or without:

commissurotomy of pulmonary valve
infundibulectomy
outflow tract prosthesis
patch graft of outflow tract
prosthetic tube for pulmonary artery
repair of ventricular septal defect (with prosthesis)
take-down of previous systemic-pulmonary artery anastomosis

35.82 Total repair of total anomalous pulmonary venous connection

One-stage total correction of total anomalous pulmonary venous connection with or without:

anastomosis between (horizontal) common pulmonary trunk and posterior wall of left atrium (side-to-side)
enlargement of foramen ovale
incision [excision] of common wall between posterior left atrium and coronary sinus and roofing of resultant defect with patch graft (synthetic)

- ligation of venous connection (descending anomalous vein) (to left innominate vein)
(to superior vena cava)
 - repair of atrial septal defect (with prosthesis)
- 35.83 Total repair of truncus arteriosus
 - One-stage total correction of truncus arteriosus with or without:
construction (with aortic homograft) (with prosthesis) of a pulmonary artery placed
from right ventricle to arteries supplying the lung
 - ligation of connections between aorta and pulmonary artery
 - repair of ventricular septal defect (with prosthesis)
- 35.84 Total correction of transposition of great vessels, not elsewhere classified
 - Arterial switch operation [Jatene]
 - Total correction of transposition of great arteries at the arterial level by switching the
great arteries, including the left or both coronary arteries, implanted in the wall
of the pulmonary artery

Excludes: baffle operation [Mustard] [Senning] ([35.91](#))
creation of shunt between right ventricle and pulmonary artery [Rastelli] ([35.92](#))
- 35.9 Other operations on valves and septa of heart

Code also cardiopulmonary bypass, if performed [extracorporeal circulation] [heart-lung machine] ([39.61](#))

 - 35.91 Interatrial transposition of venous return
 - Baffle:
 - atrial
 - interatrial
 - Mustard's operation
 - Resection of atrial septum and insertion of patch to direct systemic venous return to
tricuspid valve and pulmonary venous return to mitral valve
 - 35.92 Creation of conduit between right ventricle and pulmonary artery
 - Creation of shunt between right ventricle and (distal) pulmonary artery
 - [Rastelli's Procedure](#)

Excludes: that associated with total repair of truncus arteriosus ([35.83](#))
 - 35.93 Creation of conduit between left ventricle and aorta
 - Creation of apicoaortic shunt
 - Shunt between apex of left ventricle and aorta
 - 35.94 Creation of conduit between atrium and pulmonary artery
 - [Fontan's procedure](#)
 - [Norwood's Procedure](#)
 - 35.95 Revision of corrective procedure on heart
 - Replacement of prosthetic heart valve poppet
 - Resuture of prosthesis of:
 - septum
 - valve

Excludes: complete revision -- code to specific procedure
replacement of prosthesis or graft of:
septum ([35.50-35.63](#))
valve ([35.20-35.28](#))

35.96 Percutaneous balloon valvuloplasty
 Balloon dilation of valve
[Ballon Aorta Valvulopasty](#)
[Ballon Mitral Valvuloplasty](#)
[Ballon Pulmonal Valvuloplasty](#)
[Transcatheter Pulmonary Valvotomy & Balloon Dilation](#)

Excludes: mitral valve repair with implant ([35.97](#))

35.97 Percutaneous mitral valve repair with implant
 Endovascular mitral valve repair
 Implantation of mitral valve leaflet clip
 Transcatheter mitral valve repair

Code also any **transesophageal echocardiography [TEE] (IM) (88.720)**

Excludes: percutaneous balloon valvuloplasty ([35.96](#))

35.98 Other operations on septa of heart

35.99 Other operations on valves of heart

36	Operations on vessels of heart
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Includes: sternotomy (median) (transverse) as operative approach
 thoracotomy as operative approach

Code also any
 cardiopulmonary bypass, if performed [extracorporeal circulation] [heart-lung machine] ([39.61](#))
 injection or infusion of platelet inhibitor ([99.20](#))
 injection or infusion of thrombolytic agent ([99.10](#))

36.0 Removal of coronary artery obstruction and insertion of stent(s)

36.03 Open chest coronary artery angioplasty
 Coronary (artery):
 endarterectomy (with patch graft)
 thromboendarterectomy (with patch graft)
 Open surgery for direct relief of coronary artery obstruction

Excludes: that with coronary artery bypass graft ([36.10-36.19](#))

Code also any:
 insertion of drug-eluting coronary stent(s) ([36.07](#))
 insertion of non-drug-eluting coronary stent(s) ([36.06](#))
 number of vascular stents inserted ([00.45-00.48](#))
 number of vessels treated ([00.40-00.43](#))
 procedure on vessel bifurcation ([00.44](#))

36.04 Intracoronary artery thrombolytic infusion
 That by direct coronary artery injection, infusion, or catheterization
 Enzyme infusion
 Platelet inhibitor

Excludes: infusion of platelet inhibitor ([99.20](#))
 infusion of thrombolytic agent (99.10)
 that associated with any procedure in ([36.03](#))

36.05 Insertion Of Drug eluting or coated Balloon (IM)

36.06 Insertion of non-drug-eluting coronary artery stent(s)

Bare stent(s)
Bonded stent(s)
Drug-coated stent(s), i.e. heparin coated
Endograft(s)
Endovascular graft(s)
Stent grafts

Code also any:

number of vascular stents inserted ([00.45-00.48](#))
number of vessels treated ([00.40-00.43](#))
open chest coronary artery angioplasty ([36.03](#))
percutaneous transluminal coronary angioplasty [PTCA] or coronary atherectomy
([00.66](#))
procedure on vessel bifurcation ([00.44](#))

Excludes: insertion of drug-eluting coronary artery stent(s) ([36.07](#))

36.07 Insertion of drug-eluting coronary artery stent(s)

Endograft(s)
Endovascular graft(s)
Stent graft(s)
[Complex chronic total occlusion procedures antegrade approached](#)
[Complex chronic total occlusion procedures retrograde approached](#)
[Coronary bifurcation procedures](#)

Code also any:

number of vascular stents inserted ([00.45-00.48](#))
number of vessels treated ([00.40-00.43](#))
open chest coronary artery angioplasty ([36.03](#))
percutaneous transluminal coronary angioplasty [PTCA] or coronary atherectomy
([00.66](#))
procedure on vessel bifurcation ([00.44](#))

Excludes: drug-coated stents, e.g., heparin coated ([36.06](#))
insertion of non-drug-eluting coronary artery stent(s) ([36.06](#))

36.09 Other removal of coronary artery obstruction

Coronary angioplasty NOS

Code also any:

number of vascular stents inserted ([00.45-00.48](#))
number of vessels treated ([00.40-00.43](#))
procedure on vessel bifurcation ([00.44](#))

Excludes: that by open angioplasty ([36.03](#))
that by percutaneous transluminal coronary angioplasty [PTCA] or coronary
atherectomy ([00.66](#))

36.1 Bypass anastomosis for heart revascularization

Note: Do not assign codes from series 00.40-00.43 with codes from series ([36.10-36.19](#))

Code also : cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))
pressurized treatment of venous bypass graft [conduit] with pharmaceutical substance, if
performed ([00.16](#))

- 36.10 Aortocoronary bypass for heart revascularization, not otherwise specified
 - Direct revascularization:
 - cardiac with catheter stent, prosthesis, or vein graft
 - coronary with catheter stent, prosthesis, or vein graft
 - heart muscle with catheter stent, prosthesis, or vein graft
 - myocardial with catheter stent, prosthesis, or vein graft
 - Heart revascularization NOS
- 36.100 Aortocoronary bypass for heart revascularization with surgical ventricular restoration ^(IM)
- 36.101 Aortocoronary bypass for heart revascularization with IVS Rupture Repair ^(IM)
- 36.11 (Aorto)coronary bypass of one coronary artery
- 36.12 (Aorto)coronary bypass of two coronary arteries
- 36.13 (Aorto)coronary bypass of three coronary arteries
- 36.14 (Aorto)coronary bypass of four or more coronary arteries
- 36.15 Single internal mammary-coronary artery bypass
 - Anastomosis (single):
 - mammary artery to coronary artery
 - thoracic artery to coronary artery
- 36.16 Double internal mammary-coronary artery bypass
 - Anastomosis, double:
 - mammary artery to coronary artery
 - thoracic artery to coronary artery
- 36.17 Abdominal - coronary artery bypass
 - Anastomosis:
 - Gastroepiploic - coronary artery
- 36.18 Aortocoronary bypass for heart revascularization with valve procedure ^(IM)
- 36.19 Other bypass anastomosis for heart revascularization
- 36.2 Heart revascularization by arterial implant
 - Implantation of:
 - aortic branches [ascending aortic branches] into heart muscle
 - blood vessels into myocardium
 - internal mammary artery [internal thoracic artery] into:
 - heart muscle
 - myocardium
 - ventricle
 - ventricular wall
 - Indirect heart revascularization NOS
 - Surgical procedure of the Aortic Root with bypass anastomosis for heart revascularization
 - Extracorporeal Shockwave Myocardial Revascularization (ESMR)
- 36.3 Other heart revascularization
 - 36.31 Open chest transmyocardial revascularization
 - 36.32 Other transmyocardial revascularization
 - 36.33 Endoscopic transmyocardial revascularization
 - Thoracoscopic transmyocardial revascularization

- 36.34 Percutaneous transmyocardial revascularization
Endovascular transmyocardial revascularization
 - 36.39 Other heart revascularization
 - Abrasion of epicardium
 - Cardio-omentopexy
 - Intrapericardial poudrage
 - Myocardial graft:
 - mediastinal fat
 - omentum
 - pectoral muscles
 - 36.9 Other operations on vessels of heart
- Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))
- 36.91 Repair of aneurysm of coronary vessel
 - 36.99 Other operations on vessels of heart
 - Exploration of coronary artery
 - Incision of coronary artery
 - Ligation of coronary artery
 - Repair of arteriovenous fistula

37	Other operations on heart and pericardium
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Code also any injection or infusion of platelet inhibitor ([99.20](#))

37.0 Pericardiocentesis

37.1 Cardiotomy and pericardiotomy

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))

37.10 Incision of heart, not otherwise specified
Cardiolysis NOS

37.11 Cardiotomy
Incision of:
atrium
endocardium
myocardium
ventricle

37.12 Pericardiotomy
Pericardial window operation
Pericardiolysis
Pericardiotomy

37.2 Diagnostic procedures on heart and pericardium

37.20 Noninvasive programmed electrical stimulation (NIPS)

Excludes: that as part of intraoperative testing - omit code
catheter based invasive electrophysiologic testing ([37.26](#))
device interrogation only without arrhythmia induction (bedside check) ([89.45-89.49](#))

37.21 Right heart cardiac catheterization
Cardiac catheterization NOS

Excludes: that with catheterization of left heart ([37.23](#))

37.22 Left heart cardiac catheterization

Excludes: that with catheterization of right heart ([37.23](#))

37.23 Combined right and left heart cardiac catheterization

37.24 Biopsy of pericardium

37.25 Biopsy of heart

[Biopsy of heart with general anaesthesia](#)

37.26 Catheter based invasive electrophysiologic testing
Electrophysiologic studies [EPS]

Code also any concomitant procedure

Excludes: that as part of intraoperative testing - omit code
device interrogation only without arrhythmia induction (bedside check) ([89.45-89.49](#))
His bundle recording ([37.29](#))
noninvasive programmed electrical stimulation (NIPS) ([37.20](#))

- 37.27 Cardiac mapping
 Cardiac mapping, with general anaesthesia
 Other nonoperative cardiac diagnostic procedure (Ajmaline)

Code also any concomitant procedure

Excludes: electrocardiogram ([89.52](#))
 His bundle recording ([37.29](#))

- 37.28 Intracardiac echocardiography
 Echocardiography of heart chambers
 ICE

Code also any synchronous Doppler flow mapping ([88.72](#))

Excludes: intravascular imaging of coronary vessels (intravascular ultrasound)(IVUS) ([00.24](#))

- 37.29 Other diagnostic procedures on heart and pericardium

Excludes: angiocardiology ([88.50-88.58](#))
 cardiac function tests ([89.41-89.69](#))
 cardiovascular radioisotopic scan and function study ([92.05](#))
 coronary arteriography ([88.55-88.57](#))
 diagnostic pericardiocentesis ([37.0](#))
 diagnostic ultrasound of heart ([88.72](#))
 x-ray of heart ([87.49](#))

- 37.3 Pericardiectomy and excision of lesion of heart

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine], if performed ([39.61](#))

- 37.31 Pericardiectomy

Excision of:

 adhesions of pericardium constricting scar of:
 epicardium
 pericardium

- 37.32 Excision of aneurysm of heart
 Repair of aneurysm of heart

- 37.33 Excision or destruction of other lesion or tissue of heart, open approach
 Ablation or incision of heart tissue (cryoablation) (electrocurrent) (laser) (microwave)
 (radiofrequency) (resection) (ultrasound), open chest approach
 Cox-maze procedure
 Maze procedure
 That by median sternotomy
 That by thoracotomy without use of thoracoscope
 Surgical Ablation for Atrial Fibrillation (MAZE)

Excludes: ablation, excision or destruction of lesion or tissue of heart:
 endovascular approach ([37.34](#))
 thoracoscopic approach ([37.37](#))
 excision or destruction of left atrial appendage (LAA) ([37.36](#))

- 37.34 Excision or destruction of other lesion or tissue of heart, endovascular approach
 Ablation of heart tissue (cryoablation) (electrocurrent) (laser) (microwave)
 (radiofrequency) (ultrasound), via peripherally inserted catheter
 Modified maze procedure, percutaneous approach
 Endovenous ablation :
 Endovenous Ablation for Arteriovenous Malformation
 Endovenous Ablation for Varicose Veins

37.340 Catheter based invasive electrophysiologic 3-dimensional cryo-ablation
Catheter based invasive electrophysiologic 3-dimensional cryo-ablation with general anaesthesia

37.341 Catheter based invasive electrophysiologic 3-dimensional radiofrequency ablation

Catheter based invasive electrophysiologic 3-dimensional radiofrequency ablation with general anaesthesia

37.242 Catheter based invasive electrophysiologic cryo-ablation

Catheter based invasive electrophysiologic cryo-ablation with general anaesthesia

Excludes: ablation, excision or destruction of lesion or tissue of heart:

open approach ([37.33](#))

thoroscopic approach ([37.37](#))

37.35 Partial ventriculectomy

Ventricular reduction surgery

Ventricular remodeling

Code also any synchronous: mitral valve repair ([35.02](#), [35.12](#))

mitral valve replacement ([35.23-35.24](#))

37.36 Excision or destruction of left atrial appendage (LAA)

Includes: Thoracoscopic approach, minithoracotomy approach

Clipping of left atrial appendage

Exclusion of left atrial appendage

Oversewing of left atrial appendage

Stapling of left atrial appendage

Code also any concomitant procedure performed

Excludes: ablation, excision or destruction of lesion or tissue of heart, endovascular approach ([37.34](#))

37.37 Excision or destruction of other lesion or tissue of heart, thoracoscopic approach

Ablation or incision of heart tissue (cryoablation) (electrocautery) (laser) (microwave) (radiofrequency) (resection) (ultrasound), via thoracoscope

Modified maze procedure, thoracoscopic approach

That via thoroscopically-assisted approach (without thoracotomy) (with port access) (with sub-xiphoid incision)

Excludes: ablation, excision or destruction of lesion or tissue of heart:

endovascular approach ([37.34](#))

open approach ([37.33](#))

thoroscopic excision or destruction of left atrial appendage [LAA] ([37.36](#))

37.4 Repair of heart and pericardium

37.41 Implantation of prosthetic cardiac support device around the heart

Cardiac support device (CSD)

Epicardial support device

Fabric (textile) (mesh) device

Ventricular support device on surface of heart

Code also any:

cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] if performed ([39.61](#))

mitral valve repair ([35.02](#), [35.12](#))

mitral valve replacement ([35.23-35.24](#))

transesophageal echocardiography ([88.72](#))

Excludes:

circulatory assist systems ([37.61-37.68](#))

- 37.49 Other repair of heart and pericardium
- 37.5 Heart replacement procedures
- 37.51 Heart transplantation
Excludes: combined heart-lung transplantation ([33.6](#))
- 37.52 Implantation of total internal biventricular heart replacement system
Artificial heart
Note: This procedure includes substantial removal of part or all of the biological heart. Both ventricles are resected, and the native heart is no longer intact. Ventriculectomy is included in this procedure; do not code separately.
Excludes: implantation of heart assist system [VAD] ([37.62](#), [37.65](#), [37.66](#), [37.68](#))
- 37.53 Replacement or repair of thoracic unit of (total) replacement heart system
Excludes: replacement and repair of heart assist system [VAD]([37.63](#))
- 37.54 Replacement or repair of other implantable component of (total) replacement heart system
Implantable battery
Implantable controller
Transcutaneous energy transfer [TET] device
Excludes: replacement and repair of heart assist system [VAD] ([37.63](#))
replacement or repair of thoracic unit of (total) replacement heart system ([37.53](#))
- 37.55 Removal of internal biventricular heart replacement system
Explantation of artificial heart
Code also any concomitant procedure, such as:
combined heart-lung transplantation ([33.6](#))
heart transplantation ([37.51](#))
implantation of internal biventricular heart replacement system ([37.52](#))
Excludes: explantation [removal] of external heart assist system ([37.64](#))
explantation [removal] of percutaneous external heart assist device ([97.44](#))
nonoperative removal of heart assist system ([97.44](#))
that with replacement or repair of heart replacement system ([37.53](#), [37.54](#))
- 37.6 Implantation of heart and circulatory assist system(s)
Excludes: implantation of prosthetic cardiac support system ([37.41](#))
- 37.60 Implantation or insertion of biventricular external heart assist system
Temporary cardiac support for both left and right ventricles, inserted in the same operative episode
Includes: open chest (sternotomy) procedure for cannulae attachments
Note: Device (outside the body but connected to heart) with external circulation pump. Ventriculotomy is included; do not code separately.
Excludes: implantation of internal biventricular heart replacement system (artificial heart) ([37.52](#))
implant of pulsation balloon ([37.61](#))
insertion of percutaneous external heart assist device ([37.68](#))
insertion of temporary non-implantable extracorporeal circulatory assist device ([37.62](#))
- 37.61 Implant of pulsation balloon
- 37.62 Insertion of temporary non-implantable extracorporeal circulatory assist device
Insertion of heart assist system, NOS
Insertion of heart pump
Excludes: implantation of total internal biventricular heart replacement system [artificial heart] ([37.52](#))

implant of external heart assist system ([37.65](#))
 insertion of implantable extracorporeal heart assist system ([37.66](#))
 insertion of percutaneous external heart assist device ([37.68](#))
 removal of heart assist system ([37.64](#))

Note: Includes: explantation [37.80-37.87](#)

of this device; do not code separately.

37.63 Repair of heart assist system

Replacement of parts of an existing ventricular assist device (VAD)

Excludes: replacement or repair of other implantable component of (total) replacement heart system [artificial heart] ([37.54](#))
 replacement or repair of thoracic unit of (total) replacement heart system [artificial heart] ([37.53](#))

37.64 Removal of external heart assist system(s) or device(s)

Explantation of external device(s) providing left and right ventricular support
 Explantation of single external device and cannulae

Excludes: explantation [removal] of percutaneous external heart assist device ([97.44](#))
 nonoperative removal of heart assist system ([97.44](#))
 temporary non-implantable extracorporeal circulatory assist device ([37.62](#))
 that with replacement of implant ([37.63](#))

37.65 Implant of single ventricular (extracorporeal) external heart assist system
 Insertion of one device into one ventricle

Note: Device (outside the body but connected to heart) with external circulation and pump.

Note: Insertion or implantation of one external VAD for left or right heart support.

Includes: open chest (sternotomy) procedure for cannulae attachments

Excludes: implant of pulsation balloon ([37.61](#))
 implantation of total internal biventricular heart replacement system ([37.52](#))
 insertion of implantable heart assist system ([37.66](#))
 insertion of percutaneous external heart assist device ([37.68](#))
 insertion or implantation of two external VADs for simultaneous right and left heart support ([37.60](#))
 that without sternotomy ([37.62](#))

37.66 Insertion of implantable heart assist system

Axial flow heart assist system
 Diagonal pump heart assist system
 Left ventricular assist device (LVAD)
 Pulsatile heart assist system
 Right ventricular assist device (RVAD)
 Rotary pump heart assist system
 Transportable, implantable heart assist system.
 Ventricular assist device (VAD) not otherwise specified

Note: Device directly connected to the heart and implanted in the upper left quadrant of peritoneal cavity.

Note: This device can be used for either destination therapy (DT) or bridge-to-transplant (BTT)

Excludes: implant of pulsation balloon ([37.61](#))
 implantation of total internal biventricular heart replacement system [artificial heart] ([37.52](#))
 insertion of percutaneous external heart assist device ([37.68](#))

37.67 Implantation of cardiomyostimulation system

Note: Two-step open procedure consisting of transfer of one end of the latissimus dorsi muscle; wrapping it around the heart; rib resection; implantation of epicardial cardiac pacing leads into the right ventricle; tunneling and pocket creation for the cardiomyostimulator.

37.68 Insertion of percutaneous external heart assist device

Includes: percutaneous [femoral] insertion of cannulae attachments
Circulatory assist device
Extrinsic heart assist device
pVAD
Percutaneous heart assist device

37.7 Insertion, revision, replacement, and removal of leads; insertion of temporary pacemaker system; or revision of cardiac device pocket

Code also any insertion and replacement of pacemaker device ([37.80-37.87](#))

Excludes: implantation or replacement of transvenous lead [electrode] into left ventricular cardiac venous system ([00.52](#))

37.70 Initial insertion of lead [electrode], not otherwise specified

[Initial insertion of leadless pacemaker](#)

[Initial insertion of leadless pacemaker, with general anaesthesia](#)

Excludes: insertion of temporary transvenous pacemaker system ([37.78](#))
replacement of atrial and/or ventricular lead(s) ([37.76](#))

37.71 Initial insertion of transvenous lead [electrode] into ventricle

Excludes: insertion of temporary transvenous pacemaker system ([37.78](#))
replacement of atrial and/or ventricular lead(s) ([37.76](#))

37.72 Initial insertion of transvenous leads [electrodes] into atrium and ventricle

Excludes: insertion of temporary transvenous pacemaker system ([37.78](#))
replacement of atrial and/or ventricular lead(s) ([37.76](#))

37.73 Initial insertion of transvenous lead [electrode] into atrium

Excludes: insertion of temporary transvenous pacemaker system ([37.78](#))
replacement of atrial and/or ventricular lead(s) ([37.76](#))

37.74 Insertion or replacement of epicardial lead [electrode] into epicardium

Insertion or replacement of epicardial by:
sternotomy
thoracotomy

Excludes: replacement of atrial and/or ventricular lead(s) ([37.76](#))

37.75 Revision of lead [electrode]

Repair of electrode [removal with re-insertion]
Repositioning of lead(s) (AICD) (cardiac device) (CRT-D) (CRT-P) (defibrillator)
(pacemaker) (pacing) (sensing) [electrode]
Revision of lead NOS

Excludes: repositioning of temporary transvenous pacemaker system -- omit code

37.76 Replacement of transvenous atrial and/or ventricular lead(s) [electrode]

Removal or abandonment of existing transvenous or epicardial lead(s) with
transvenous lead(s) replacement

[Removal of lead\(s\) \(electrodes\) with replacement](#)

[Removal of lead\(s\) \(electrodes\) with replacement and general anaesthesia](#)

Excludes: replacement of epicardial lead [electrode] ([37.74](#))

37.77 Removal of lead(s) [electrode] without replacement

Removal:

epicardial lead (transthoracic approach)

transvenous lead(s)

[lead\(s\) \(electrodes\) without replacement, with general anaesthesia](#)

Excludes: removal of temporary transvenous pacemaker system -- omit code that with replacement of:

atrial and/or ventricular lead(s) [electrode] ([37.76](#))

epicardial lead [electrode] ([37.74](#))

37.78 Insertion of temporary transvenous pacemaker system ([TPM](#))

[TPM in neonatus](#)

[TPM in pediatrik](#)

Excludes: intraoperative cardiac pacemaker ([39.64](#))

37.79 Revision or relocation of cardiac device pocket

Debridement and reforming pocket (skin and subcutaneous tissue)

Insertion of loop recorder

Relocation of pocket [creation of new pocket] pacemaker or CRT-P

Removal of cardiac device/pulse generator without replacement

Removal of the implantable hemodynamic pressure sensor (lead) and monitor device

Removal without replacement of cardiac resynchronization defibrillator device

Repositioning of implantable hemodynamic pressure sensor (lead) and monitor device

Repositioning of pulse generator

Revision of cardioverter/defibrillator (automatic) pocket

Revision of pocket for intracardiac hemodynamic monitoring

Revision or relocation of CRT-D pocket

Revision or relocation of pacemaker, defibrillator, or other implanted cardiac device pocket

[Insertion of implantable loop recorder \(ILR\)](#)

[Insertion of implantable loop recorder \(ILR\), with general anaesthesia](#)

[revision or relocation of cardiac device procedure include ILR](#)

[Revision or relocation of pacemaker pocket, with general anaesthesia](#)

[Revision or removal of pacemaker device, with general anaesthesia](#)

Excludes: removal of loop recorder ([86.05](#))

37.8 Insertion, replacement, removal, and revision of pacemaker device

Note: Device testing during procedure - omit code

Code also any lead insertion, lead replacement, lead removal and/or lead revision ([37.70-37.77](#))

Excludes: implantation of cardiac resynchronization pacemaker [CRT-P] ([00.50](#))

implantation or replacement of cardiac resynchronization pacemaker pulse generator only [CRT-P] ([00.53](#))

37.80 Insertion of permanent pacemaker, initial or replacement, type of device not specified

37.81 Initial insertion of single-chamber device, not specified as rate responsive

[Initial insertion of single chamber leadless](#)

Excludes: replacement of existing pacemaker device ([37.85-37.87](#))

- 37.82 Initial insertion of single-chamber device, rate responsive
 Rate responsive to physiologic stimuli other than atrial rate
[Initial insertion of single chamber pacemaker , MRI](#)
[Initial insertion of single-chamber pacemaker, rate-responsive, MRI-level](#)
[Initial insertion of single-chamber pacemaker, rate-responsive, MRI-level with general anaesthesia](#)
[Initial insertion of single-chamber pacemaker, rate-responsive, with general anaesthesia](#)

Excludes: replacement of existing pacemaker device ([37.85-37.87](#))

- 37.83 Initial insertion of dual-chamber device
 Atrial ventricular sequential device
[Initial insertion of dual-chamber device \(Initial insertion of dual-chamber pacemaker, with general anaesthesia\)](#)
[Initial insertion dual chamber pacemaker, MRI](#)
[Initial insertion of dual-chamber pacemaker, MRI-level](#)
[Initial insertion of dual-chamber pacemaker, MRI-level, with general anaesthesia](#)

Excludes: replacement of existing pacemaker device ([37.85-37.87](#))

- 37.85 Replacement of any type pacemaker device with single-chamber device, not specified as rate responsive

[Replacement of any type of pacemaker device with single-chamber device, rate responsive, with general anaesthesia](#)

- 37.86 Replacement of any type of pacemaker device with single-chamber device, rate responsive
 Rate responsive to physiologic stimuli other than atrial rate

- 37.87 Replacement of any type pacemaker device with dual-chamber device
 Atrial ventricular sequential device
[Replacement of any type of pacemaker device with dual-chamber device, MRI-level](#)
[Replacement of any type of pacemaker device with dual-chamber device, MRI-level, with general anaesthesia](#)
[Replacement of any type of pacemaker device with dual-chamber device, MRI-level, with general anaesthesia](#)

- 37.89 Revision or removal of pacemaker device
 Removal without replacement of cardiac resynchronization pacemaker device [CRT-P]
 Repair of pacemaker device

Excludes: removal of temporary transvenous pacemaker system -- omit code
 replacement of existing pacemaker device ([37.85-37.87](#))
 replacement of existing pacemaker device with CRT-P pacemaker device ([00.53](#))

37.9 Other operations on heart and pericardium

37.90 Insertion of left atrial appendage device

Left atrial filter

Left atrial occluder

Transseptal catheter technique

[Insertion of left atrial appendage device, with general anaesthesia](#)

[Insertion of left atrial appendage device, with transesophageal echocardiography](#)

37.91 Open chest cardiac massage

Excludes: closed chest cardiac massage (99.63)

37.92 Injection of therapeutic substance into heart

37.93 Injection of therapeutic substance into pericardium

37.94 Implantation or replacement of automatic cardioverter/defibrillator, total system [AICD]

Implantation of defibrillator with leads (epicardial patches), formation of pocket (abdominal fascia) (subcutaneous), any transvenous leads, intraoperative procedures for evaluation of lead signals, and obtaining defibrillator threshold measurements

Techniques:

lateral thoracotomy

medial sternotomy

subxiphoid procedure

[Implantation of automatic cardioverter/defibrillator pulse generator only, with general anaesthesia](#)

[Implantation of dual chamber cardioverter/defibrillator](#)

[Implantation of subcutaneous cardioverter/defibrillator](#)

[Implantation of subcutaneous cardioverter/defibrillator, with general anaesthesia](#)

[Implantation or replacement of automatic cardioverter/ defibrillator, total system \(AICD\), with general](#)

[initial insertion of dual chamber ICD *](#)

[initial insertion of subcutaneous defibrillator ICD *](#)

[Implantation or replacement of transvenous lead \(electrode\) into left ventricular coronary venous system, with general anaesthesia](#)

Note: Device testing during procedure - omit code

Code also extracorporeal circulation, if performed ([39.61](#))

Code also any concomitant procedure [e.g., coronary bypass ([36.10-36.19](#)) or CCM, total system ([17.51](#))]

Excludes: implantation of cardiac resynchronization defibrillator, total system [CRT-D] ([00.51](#))

37.95 Implantation of automatic cardioverter/defibrillator lead(s) only

37.96 Implantation of automatic cardioverter/defibrillator pulse generator only

[Implantation of dual chamber cardioverter/defibrillator, with general anaesthesia](#)

Note: Device testing during procedure - omit code

Excludes: implantation or replacement of cardiac resynchronization defibrillator, pulse generator device only [CRT-D] ([00.54](#))

37.97 Replacement of automatic cardioverter/defibrillator lead(s) only

Excludes: replacement of epicardial lead [electrode] into epicardium ([37.74](#))

replacement of transvenous lead [electrode] into left ventricular coronary venous system ([00.52](#))

37.98 Replacement of automatic cardioverter/defibrillator pulse generator only

Note: Device testing during procedure - omit code

Excludes: replacement of cardiac resynchronization defibrillator, pulse generator device only [CRT-D] ([00.54](#))

37.99 Other

Excludes: cardiac retraining ([93.36](#))

conversion of cardiac rhythm ([99.60-99.69](#))

implantation of prosthetic cardiac support device ([37.41](#))

insertion of left atrial appendage device ([37.90](#))

maze procedure (Cox-maze), open ([37.33](#))

maze procedure, endovascular approach ([37.34](#))

repositioning of pulse generator ([37.79](#))

revision of lead(s) ([37.75](#))

revision or relocation of pacemaker, defibrillator or other implanted cardiac device pocket ([37.79](#))

38	Incision, excision, and occlusion of vessels
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Code also

any application or administration of an adhesion barrier substance ([99.77](#))

cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))

Excludes: that of coronary vessels ([00.66](#), [36.03](#), [36.04](#), [36.09](#), [36.10-36.99](#))

The following fourth-digit subclassification is for use with appropriate categories in section 38.0, 38.1, 38.3, 38.5, 38.6, 38.8, and 38.9 according to site. Valid fourth-digits are in [brackets] at the end of each code/description.

- 0 unspecified site
- 1 intracranial vessels
 - Cerebral (anterior) (middle)
 - Circle of Willis
 - Posterior communicating artery
- 2 other vessels of head and neck
 - Carotid artery (common) (external) (internal)
 - Jugular vein (external) (internal)
- 3 upper limb vessels
 - Axillary
 - Brachial
 - Radial
 - Ulnar
- 4 aorta
- 5 other thoracic vessels
 - Innominate
 - Pulmonary (artery) (vein)
 - Subclavian
 - Vena cava, superior

- 6 abdominal arteries
 - Celiac
 - Gastric
 - Hepatic
 - Iliac
 - Mesenteric
 - Renal
 - Splenic
 - Umbilical
- Excludes: abdominal aorta (4)
- 7 abdominal veins
 - Iliac
 - Portal
 - Renal
 - Splenic
 - Vena cava (inferior)
- 8 lower limb arteries
 - Femoral (common) (superficial)
 - Popliteal
 - Tibial
- 9 lower limb veins
 - Femoral
 - Popliteal
 - Saphenous
 - Tibial
- 38.0 Incision of vessel
[0-9]
 - Embolectomy
 - Thrombectomy
 - Hybrid Procedure
 - Intraarterial thrombolysis
 - Hybrid phlebectomy and endovenous ablation
- Excludes: endovascular removal of obstruction from head and neck vessel(s) ([39.74](#))
puncture or catheterization of any:
 - artery ([38.91](#), [38.98](#))
 - vein ([38.92-38.95](#), [38.99](#))
- 38.1 Endarterectomy
[0-6,8]
 - Endarterectomy with:
 - embolectomy
 - patch graft
 - temporary bypass during procedure
 - thrombectomy
- Code also any:
 - number of vascular stents inserted ([00.45-00.48](#))
 - number of vessels treated ([00.40-00.43](#))
 - procedure on vessel bifurcation ([00.44](#))
- 38.2 Diagnostic procedures on blood vessels
 - Excludes: adjunct vascular system procedures ([00.40-00.43](#))
 - 38.21 Biopsy of blood vessel

38.22 Percutaneous angiography

Excludes: angiography of eye ([95.12](#))

38.23 Intravascular spectroscopy

Includes: spectroscopy of both coronary and peripheral vessels

Intravascular chemography

Near infrared (NIR) spectroscopy

Excludes: intravascular imaging of:

coronary vessels ([00.24](#), [38.24](#))

peripheral vessels ([00.23](#), [38.25](#))

38.24 Intravascular imaging of coronary vessel(s) by optical coherence tomography [OCT]

38.25 Intravascular imaging of non-coronary vessel(s) by optical coherence tomography [OCT]

Excludes: intravascular imaging of coronary vessel(s) by OCT ([38.24](#))

38.29 Other diagnostic procedures on blood vessels

Excludes: blood vessel thermography ([88.86](#))

circulatory monitoring ([89.61-89.69](#))

contrast:

angiocardiology ([88.50-88.58](#))

arteriography ([88.40-88.49](#))

phlebography ([88.60-88.67](#))

impedance phlebography ([88.68](#))

peripheral vascular ultrasonography ([88.77](#))

plethysmogram ([89.58](#))

38.3 Resection of vessel with anastomosis
[0-9]

Angiectomy

Excision of:

aneurysm (arteriovenous) with anastomosis

blood vessel (lesion) with anastomosis

38.4 Resection of vessel with replacement
[0-9]

Angiectomy

Excision of:

aneurysm (arteriovenous) or blood vessel (lesion) with replacement

Partial resection with replacement

Excludes: endovascular repair of aneurysm ([39.71-39.79](#))

Requires the use of one of the following fourth-digit subclassifications to identify site:

- 0 unspecified site
- 1 intracranial vessels
 - Cerebral (anterior) (middle)
 - Circle of Willis
 - Posterior communicating artery
- 2 other vessels of head and neck
 - Carotid artery (common) (external) (internal)
 - Jugular vein (external) (internal)
- 3 upper limb vessels
 - Axillary
 - Brachial

- Radial
- Ulnar
- 4 aorta, abdominal
- Code also any thoracic vessel involvement (thoracoabdominal procedure) ([38.45](#))
- 5 thoracic vessels
 - Aorta (thoracic)
 - Innominate
 - Pulmonary (artery) (vein)
 - Subclavian
 - Vena cava, superior
- Code also any abdominal aorta involvement (thoracoabdominal procedure) ([38.44](#))
- 6 abdominal arteries
 - Celiac
 - Gastric
 - Hepatic
 - Iliac
 - Mesenteric
 - Renal
 - Splenic
 - Umbilical
- Excludes: abdominal aorta (4)
- 7 abdominal veins
 - Iliac
 - Portal
 - Renal
 - Splenic
 - Vena cava (inferior)
- 8 lower limb arteries
 - Femoral (common) (superficial)
 - Popliteal
 - Tibial
- 9 lower limb veins
 - Femoral
 - Popliteal
 - Saphenous
 - Tibial
- 38.5 Ligation and stripping of varicose veins
[0-3,5,7,9]
- Excludes: ligation of varices:
 - esophageal ([42.91](#))
 - gastric ([44.91](#))
- 38.6 Other excision of vessel
[0-9]
- Excision of blood vessel (lesion) NOS
[Phlebectomy](#)
- Excludes: excision of vessel for aortocoronary bypass ([36.10-36.14](#))
- excision with:
 - anastomosis ([38.30-38.39](#))
 - graft replacement ([38.40-38.49](#))
 - implant ([38.40-38.49](#))

38.7 Interruption of the vena cava
 Insertion of implant or sieve in vena cava
 Ligation of vena cava (inferior) (superior)
 Plication of vena cava
[Insertion of Vena Cava Filter](#)

38.8 Other surgical occlusion of vessels
 [0-9]
 Clamping of blood vessel
 Division of blood vessel
 Ligation of blood vessel
 Occlusion of blood vessel
[Percutaneous PDA occlusion](#)
[Percutaneous PDA occlusion in neonatus](#)

Excludes: adrenal vessels ([07.43](#))
 esophageal varices ([42.91](#))
 gastric or duodenal vessel for ulcer ([44.40-44.49](#))
 gastric varices ([44.91](#))
 meningeal vessel ([02.13](#))
 percutaneous transcatheter infusion embolization ([99.29](#))
 spermatic vein for varicocele ([63.1](#))
 surgical occlusion of vena cava ([38.7](#))
 that for chemoembolization ([99.25](#))
 that for control of (postoperative) hemorrhage:
 anus ([49.95](#))
 bladder ([57.93](#))
 following vascular procedure ([39.41](#))
 nose ([21.00-21.09](#))
 prostate ([60.94](#))
 tonsil ([28.7](#))
 thyroid vessel ([06.92](#))
 transcatheter (infusion) [99.29](#)

38.850 PDA stenting/PDA closure with device (IM)

38.9 Puncture of vessel

Excludes: that for circulatory monitoring ([89.60-89.69](#))

38.91 Arterial catheterization

38.92 Umbilical vein catheterization

38.93 Venous catheterization, not elsewhere classified
[Peripheral inserted central catheter](#)

Excludes: that for cardiac catheterization ([37.21-37.23](#))
 that for renal dialysis ([38.95](#))
 that with guidance (electrocardiogram) (fluoroscopy) (ultrasound) ([38.97](#))

38.94 Venous cutdown

38.95 Venous catheterization for renal dialysis
[Insertion of hemodialysis catheter](#)

Excludes: insertion of totally implantable vascular access device [VAD] ([86.07](#))

38.97 Central venous catheter placement with guidance

Includes guidance by:
 electrocardiogram
 fluoroscopy
 ultrasound

38.98 Other puncture of artery

Excludes: that for:

arteriography ([88.40-88.49](#))
 coronary arteriography ([88.55-88.57](#))

38.99 Other puncture of vein

Phlebotomy

Excludes: that for:

angiography of veins ([88.60-88.68](#))
 extracorporeal circulation ([39.61](#), [50.92](#))
 injection or infusion of:
 sclerosing solution ([39.92](#))
 therapeutic or prophylactic substance ([99.11-99.29](#))
 perfusion ([39.96-39.97](#))
 phlebography ([88.60-88.68](#))
 transfusion ([99.01-99.09](#))

39	Other operations on vessels
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Excludes: those on coronary vessels ([36.03-36.99](#))

39.0 Systemic to pulmonary artery shunt

Descending aorta-pulmonary artery anastomosis (graft)
 Left to right anastomosis (graft)
 Subclavian-pulmonary anastomosis (graft)

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))

39.1 Intra-abdominal venous shunt

Anastomosis:
 mesocaval
 portacaval
 portal vein to inferior vena cava
 splenic and renal veins
 transjugular intrahepatic portosystemic shunt (TIPS)

Excludes: peritoneovenous shunt ([54.94](#))

39.2 Other shunt or vascular bypass

Code also pressurized treatment of venous bypass graft [conduit] with pharmaceutical substance, if performed ([00.16](#))

39.21 Caval-pulmonary artery anastomosis

Code also cardiopulmonary bypass ([39.61](#))

39.22 Aorta-subclavian-carotid bypass

Bypass (arterial):
 aorta to carotid and brachial
 aorta to subclavian and carotid
 carotid to subclavian

39.23 Other intrathoracic vascular shunt or bypass
Intrathoracic (arterial) bypass graft NOS

Excludes: coronary artery bypass ([36.10-36.19](#))

39.24 Aorta-renal bypass

39.25 Aorta-iliac-femoral bypass
Bypass:
aortofemoral
aortoiliac
aortoiliac to popliteal
aortopopliteal
iliofemoral [iliac-femoral]

39.26 Other intra-abdominal vascular shunt or bypass
Bypass:
aortoceliac
aortic-superior mesenteric
common hepatic-common iliac-renal
Intra-abdominal arterial bypass graft NOS

Excludes: peritoneovenous shunt ([54.94](#))

39.27 Arteriovenostomy for renal dialysis
Anastomosis for renal dialysis
Formation of (peripheral) arteriovenous
fistula for renal [kidney] dialysis

39.270 Insertion of temporary central dialysis catheter (IM)

39.271 Insertion of permanent central dialysis catheter (IM)

Includes : Laparoscopic repair of indwelling peritoneal catheter for dialysis

39.272 Insertion of peritoneal dialysis catheter: peritoneoscopy technique (IM)

39.273 Insertion of peritoneal dialysis catheter: percutaneous technique (IM)

39.274 Insertion of peritoneal dialysis catheter: open technique (IM)

Code also any renal dialysis ([39.95](#))

39.28 Extracranial-intracranial (EC-IC) vascular bypass

39.29 Other (peripheral) vascular shunt or bypass
Bypass (graft):
axillary-brachial
axillary-femoral [axillofemoral] (superficial)
brachial
femoral-femoral
femoroperoneal
femoropopliteal (arteries)
femorotibial (anterior) (posterior)
popliteal
vascular NOS

Excludes: peritoneovenous shunt ([54.94](#))

39.3 Suture of vessel
Repair of laceration of blood vessel

Excludes: any other vascular puncture closure device - omit code
suture of aneurysm ([39.52](#))
that for control of hemorrhage (postoperative):

anus ([49.95](#))
bladder ([57.93](#))
following vascular procedure ([39.41](#))
nose ([21.00-21.09](#))
prostate ([60.94](#))
tonsil ([28.7](#))

- 39.30 Suture of unspecified blood vessel
- 39.31 Suture of artery
[Bentall's Procedure](#)
- 39.32 Suture of vein
- 39.4 Revision of vascular procedure
- 39.41 Control of hemorrhage following vascular surgery
Excludes: that for control of hemorrhage (post operative):
anus ([49.95](#))
bladder ([57.93](#))
nose ([21.00-21.09](#))
prostate ([60.94](#))
tonsil ([28.7](#))
- 39.42 Revision of arteriovenous shunt for renal dialysis
Conversion of renal dialysis:
end-to-end anastomosis to end-to-side
end-to-side anastomosis to end-to-end
vessel-to-vessel cannula to arteriovenous shunt
Removal of old arteriovenous shunt and creation of new shunt
Excludes: replacement of vessel-to-vessel cannula ([39.94](#))
- 39.43 Removal of arteriovenous shunt for renal dialysis
Excludes: that with replacement [revision] of shunt ([39.42](#))
- 39.49 Other revision of vascular procedure
Declotting (graft)
Revision of:
anastomosis of blood vessel
vascular procedure (previous)

39.5 Other repair of vessels

39.50 Angioplasty or atherectomy of other non-coronary vessel(s)

Percutaneous transluminal angioplasty (PTA) of non-coronary vessels:

Lower extremity vessels

Mesenteric artery

Renal artery

Upper extremity vessels

[Ballon Coarctation/RVOT/PA/conduit](#)

[PIAT \(Peripheral Intra Arterial Thrombolysis\)](#)

[PTA](#)

[PTV](#)

Code also any:

injection or infusion of thrombolytic agent ([99.10](#))

insertion of drug-eluting peripheral vessel stent ([00.55](#))

insertion of non-drug-eluting peripheral vessel stent(s) or stent grafts(s) ([39.90](#))

number of vascular stents inserted ([00.45-00.48](#))

number of vessels treated ([00.40-00.43](#))

procedure on vessel bifurcation ([00.44](#))

Excludes: percutaneous angioplasty or atherectomy of precerebral or cerebral vessel(s) ([00.61-00.62](#))

39.51 Clipping of aneurysm

Excludes: clipping of arteriovenous fistula ([39.53](#))

39.52 Other repair of aneurysm

Repair of aneurysm by:

coagulation

electrocoagulation

filipuncture

methyl methacrylate

suture

wiring

wrapping

Excludes: endovascular repair of aneurysm ([39.71-39.79](#))

re-entry operation (aorta) ([39.54](#))

that with:

graft replacement ([38.40-38.49](#))

resection ([38.30-38.49](#), [38.60-38.69](#))

39.53 Repair of arteriovenous fistula

Embolization of carotid cavernous fistula

Repair of arteriovenous fistula by:

clipping

coagulation

ligation and division

Excludes: repair of :

arteriovenous shunt for renal dialysis ([39.42](#))

head and neck vessels, endovascular approach ([39.72](#))

that with:

graft replacement ([38.40-38.49](#))

resection ([38.30-38.49](#), [38.60-38.69](#))

39.54 Re-entry operation (aorta)

Fenestration of dissecting aneurysm of thoracic aorta

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))

39.55 Reimplantation of aberrant renal vessel

39.56 Repair of blood vessel with tissue patch graft

Excludes: that with resection ([38.40-38.49](#))

39.57 Repair of blood vessel with synthetic patch graft

Excludes: that with resection ([38.40-38.49](#))

39.58 Repair of blood vessel with unspecified type of patch graft

Excludes: that with resection ([38.40-38.49](#))

39.59 Other repair of vessel

Aorticopulmonary window operation

Arterioplasty NOS

Construction of venous valves (peripheral)

Plication of vein (peripheral)

Reimplantation of artery

Aortic repair perkutan

39.590 Direct anastomosis for renal artery repair (IM)

39.591 Direct anastomosis for renal vein repair (IM)

39.951 Prolonged intermittent renal replacement therapy (IM)

39.952 Continuous renal replacement therapy (IM)

Code also cardiopulmonary bypass [extracorporeal circulation] [heart-lung machine] ([39.61](#))

Excludes: interruption of the vena cava ([38.7](#))

reimplantation of renal artery ([39.55](#))

that with:

graft ([39.56-39.58](#))

resection ([38.30-38.49](#), [38.60-38.69](#))

39.6 Extracorporeal circulation and procedures auxiliary to heart surgery

39.61 Extracorporeal circulation auxiliary to open heart surgery

Artificial heart and lung

Cardiopulmonary bypass

Pump oxygenator

Excludes: extracorporeal hepatic assistance ([50.92](#))

extracorporeal membrane oxygenation [ECMO] ([39.65](#))

hemodialysis (39.95)

percutaneous cardiopulmonary bypass ([39.66](#))

39.62 Hypothermia (systemic) incidental to open heart surgery

39.63 Cardioplegia

Arrest:

anoxic

circulatory

39.64 Intraoperative cardiac pacemaker

Temporary pacemaker used during and immediately following cardiac surgery

39.65 Extracorporeal membrane oxygenation [ECMO]

Excludes: extracorporeal circulation auxiliary to open heart surgery ([39.61](#))
percutaneous cardiopulmonary bypass ([39.66](#))

39.66 Percutaneous cardiopulmonary bypass
Closed chest

Excludes: extracorporeal circulation auxiliary to open heart surgery ([39.61](#))
extracorporeal hepatic assistance ([50.92](#))
extracorporeal membrane oxygenation [ECMO] ([39.65](#))
hemodialysis ([39.95](#))

39.7 Endovascular procedures on vessel(s)

Embolization
Endoluminal repair
Implantation
Occlusion
Removal
Repair

Excludes: angioplasty or atherectomy of other non-coronary vessel(s) ([39.50](#))
insertion of non-drug-eluting peripheral vessel stent(s) ([39.90](#))
other repair of aneurysm ([39.52](#))
percutaneous insertion of carotid artery stent(s) ([00.63](#))
percutaneous insertion of intracranial stent(s) ([00.65](#))
percutaneous insertion of other precerebral artery stent(s) ([00.64](#))
resection of abdominal aorta with replacement ([38.44](#))
resection of lower limb arteries with replacement ([38.48](#))
resection of thoracic aorta with replacement ([38.45](#))
resection of upper limb vessels with replacement ([38.43](#))

39.71 Endovascular implantation of graft in abdominal aorta
Endovascular repair of abdominal aortic aneurysm with graft
Stent graft(s)

Code also intra-aneurysm sac pressure monitoring (intraoperative) ([00.58](#))

39.72 Endovascular embolization or occlusion of head and neck vessels

Endograft(s)
Endovascular graft(s)
Liquid tissue adhesive (glue) embolization or occlusion
Other implant or substance for repair, embolization or occlusion
That for repair of aneurysm, arteriovenous malformation[AVM] or fistula
[Vascular embolization](#)

Excludes: embolization of head or neck vessels using bare coils ([39.75](#))
embolization of head or neck vessels using bioactive coils ([39.76](#))
mechanical thrombectomy of pre-cerebral and cerebral vessels ([39.74](#))

39.73 Endovascular implantation of graft in thoracic aorta

Endograft(s)
Endovascular graft(s)
Endovascular repair of defect of thoracic aorta with graft(s) or device(s)
Stent graft(s) or device(s)
That for repair of aneurysm, dissection, or injury

Code also intra-aneurysm sac pressure monitoring (intraoperative) ([00.58](#))

Excludes: fenestration of dissecting aneurysm of thoracic aorta ([39.54](#))

39.74 Endovascular removal of obstruction from head and neck vessel(s)
 Endovascular embolectomy
 Endovascular thrombectomy of pre-cerebral and cerebral vessels
 Mechanical embolectomy or thrombectomy
[Mechanical thrombectomy](#)

Code also: any injection or infusion of thrombolytic agent ([99.10](#))
 number of vessels treated ([00.40-00.43](#))
 procedure on vessel bifurcation ([00.44](#))

Excludes: endarterectomy of intracranial vessels and other vessels of head and neck ([38.11-38.12](#))
 occlusive endovascular embolization of head or neck vessel(s) using bare coils ([39.75](#))
 occlusive endovascular embolization of head or neck vessels(s) using bioactive coils ([39.76](#))
 open embolectomy or thrombectomy ([38.01-38.02](#))

39.75 Endovascular embolization or occlusion of vessel(s) of head or neck using bare coils
 Bare metal coils
 Bare platinum coils [BPC]
 That for treatment of aneurysm, arteriovenous malformation [AVM] or fistula

39.76 Endovascular embolization or occlusion of vessel(s) of head or neck using bioactive coils
 Biodegradable inner luminal polymer coils
 Coil embolization or occlusion utilizing bioactive coils
 Coils containing polyglycolic acid [PGA]
 That for treatment of aneurysm, arteriovenous malformation [AVM] or fistula

39.79 Other endovascular procedures on other vessels
 Endograft(s)
 Endovascular graft(s)
 Liquid tissue adhesive (glue) embolization or occlusion
 Other coil embolization or occlusion
 Other implant or substance for repair, embolization or occlusion
 Repair of aneurysm
[Coiling fistulae in one vessel](#)
[Coiling coronary fistula](#)
[Coiling MAPCA](#)
[coronary fistula intervention with coiling or alcohol ablation](#)

Excludes: abdominal aortic aneurysm resection [AAA] ([38.44](#))
 endovascular implantation of graft in abdominal aorta ([39.71](#))
 endovascular implantation of graft in thoracic aorta ([39.73](#))
 endovascular embolization or occlusion of head and neck vessels, bare metal coils ([39.75](#))
 endovascular embolization or occlusion of head and neck vessels, bioactive coils ([39.76](#))
 insertion of drug-eluting peripheral vessel stent(s) ([00.55](#))
 insertion of non-drug-eluting peripheral vessel (s) (for other than aneurysm repair) ([39.90](#))
 non-endovascular repair of arteriovenous fistula ([39.53](#))
 other surgical occlusion of vessels -see category [38.8](#)
 percutaneous transcatheter infusion ([99.29](#))
 thoracic aortic aneurysm resection ([38.45](#))
 transcatheter embolization for gastric or duodenal bleeding ([44.44](#))

39.8 Operations on carotid body, carotid sinus and other vascular bodies

Excludes: excision of glomus jugulare ([20.51](#))

39.81 Implantation or replacement of carotid sinus stimulation device, total system

Carotid sinus baroreflex activation device

Implantation of carotid sinus stimulator and lead(s)

Includes: carotid explorations

Excludes: implantation or replacement of carotid sinus stimulation lead(s) only ([39.82](#))

implantation or replacement of carotid sinus stimulation pulse generator only ([39.83](#))

39.82 Implantation or replacement of carotid sinus stimulation lead(s) only

Excludes: implantation or replacement of carotid sinus stimulation device, total system ([39.81](#))

39.83 Implantation or replacement of carotid sinus stimulation pulse generator only

Excludes: implantation or replacement of carotid sinus stimulation device, total system ([39.81](#))

39.84 Revision of carotid sinus stimulation lead(s) only

Repair of electrode [removal with re-insertion]

Repositioning of lead(s) [electrode]

39.85 Revision of carotid sinus stimulation pulse generator

Debridement and reforming pocket (skin and subcutaneous tissue)

Relocation of pocket [creation of new pocket]

Repositioning of pulse generator

Revision of carotid sinus stimulation pulse generator pocket

39.86 Removal of carotid sinus stimulation device, total system

39.87 Removal of carotid sinus stimulation lead(s) only

39.88 Removal of carotid sinus stimulation pulse generator only

39.89 Other operations on carotid body, carotid sinus and other vascular bodies

Chemodectomy

Denervation of:

aortic body

carotid body

Glomectomy, carotid

Excludes: excision of glomus jugulare ([20.51](#))

39.9 Other operations on vessels

39.90 Insertion of non-drug-eluting peripheral (non-coronary) vessel stent(s)

Bare stent(s)
 Bonded stent(s)
 Drug-coated stent(s), i.e., heparin coated
 Endograft(s)
 Endovascular graft(s)
 Endovascular recanalization techniques
 Stent graft(s)
 Ballon with stent Coarctation/RVOT/PA/conduit
 PDA stenting ([38.850 IM](#)—Hapus)

Code also any:

non-coronary angioplasty or atherectomy ([39.50](#))
 number of vascular stents inserted ([00.45-00.48](#))
 number of vessels treated ([00.40-00.43](#))
 procedure on vessel bifurcation ([00.44](#))

Excludes: insertion of drug-eluting peripheral vessel stent(s) ([00.55](#))
 percutaneous insertion of carotid artery stent(s) ([00.63](#))
 percutaneous insertion of intracranial stent(s) ([00.65](#))
 percutaneous insertion of other precerebral artery stent(s) ([00.64](#))
 that for aneurysm repair ([39.71-39.79](#))

39.91 Freeing of vessel

Dissection and freeing of adherent tissue:
 artery-vein-nerve bundle
 vascular bundle

39.92 Injection of sclerosing agent into vein

Excludes: injection:
 esophageal varices ([42.33](#))
 hemorrhoids ([49.42](#))

39.93 Insertion of vessel-to-vessel cannula

Formation of:
 arteriovenous:
 fistula by external cannula
 shunt by external cannula

Code also any renal dialysis ([39.95](#))

39.94 Replacement of vessel-to-vessel cannula

Revision of vessel-to-vessel cannula

39.95 Hemodialysis

Artificial kidney
 Hemodiafiltration
 Hemofiltration
 Renal dialysis

Excludes: peritoneal dialysis ([54.98](#))

39.951 Prolonged Intermittent Renal Replacement Therapy (IM)

39.952 Continuous Renal Replacement Therapy (IM)

39.96 Total body perfusion

Code also substance perfused ([99.21-99.29](#))

39.97 Other perfusion

Perfusion NOS

Perfusion, local [regional] of:

carotid artery

coronary artery

head

lower limb

neck

upper limb

Code also substance perfused ([99.21-99.29](#))

Excludes: perfusion of:

kidney ([55.95](#))

large intestine ([46.96](#))

liver ([50.93](#))

small intestine ([46.95](#))

SuperSaturated oxygen therapy ([00.49](#))

39.98 Control of hemorrhage, not otherwise specified

Angiotripsy

Control of postoperative hemorrhage NOS

Venotripsy

Excludes: control of hemorrhage (postoperative):

anus ([49.95](#))

bladder ([57.93](#))

following vascular procedure ([39.41](#))

nose ([21.00-21.09](#))

prostate ([60.94](#))

tonsil ([28.7](#))

that by:

ligation ([38.80-38.89](#))

suture ([39.30-39.32](#))

39.99 Other operations on vessels

Excludes: injection or infusion of therapeutic or prophylactic substance ([99.11-99.29](#))

transfusion of blood and blood components ([99.01-99.09](#))